**Computational Modeling**
Prof. Ulrich Kleinekathöfer
Spring term 2026
Project 3, due April 1, 2026 at 11:59 p.m.
to be uploaded to https://elearning.constructor.university

CONSTRUCTOR UNIVERSITY

---

### 3. Metropolis Algorithm: Sampling a Double-Well Distribution [150 points]

We aim to sample the 1D target distribution $\pi(x) \propto e^{-\beta U(x)}$ with $\beta = 1/T$ and the double-well potential $U(x) = x^4 - x^2$, in units where $k_B = 1$. Suggested temperatures are $T \in \{1.0,\ 0.5,\ 0.2\}$.
a) Implement the Metropolis algorithm in Python:

- Use a symmetric random-walk proposal $x' = x + \delta$ with $\delta$ being a uniform distributed random number in the range $[-\Delta, \Delta]$.
- Implement the Metropolis acceptance probability $P_{\text{acc}} = \min(1, \exp[-\beta(U(x') - U(x))])$.
- Run a chain and report the acceptance rate. Suggested chain length: $N_{\text{steps}} = 10^6$ (also below, you may adjust if justified). Briefly explain why detailed balance holds for a symmetric proposal.

b) Verify the sampled distribution:

- For each temperature $T \in \{1.0, 0.5, 0.2\}$, estimate $\pi(x)$ from a histogram $\hat{\pi}(x)$ of your samples (after burn-in, i.e., after neglecting initial data due to initialization effects).
- Compute a reference curve by evaluating $e^{-\beta U(x)}$ on a grid and normalizing numerically:

$$\pi_{\text{ref}}(x) = \frac{e^{-\beta U(x)}}{\int_{-\infty}^{\infty} e^{-\beta U(x)}\, dx},$$

  where you approximate the integral by numerical quadrature over a sufficiently wide interval (state your choice).

- Plot $\hat{\pi}(x)$ together with $\pi_{\text{ref}}(x)$ for each $T$ and comment on agreement/disagreement.

c) Tuning the proposal step size $\Delta$:
For each temperature, study how the choice of $\Delta$ affects sampling. Scan $\Delta$ over a reasonable range (e.g. logarithmically spaced values). Plot acceptance rate vs. $\Delta$. Choose an "optimal" $\Delta$ for each $T$ by balancing acceptance and exploration. State clearly what criterion you use.
d) Mixing, metastability, and barrier crossing:
Define the "well index" $w_t = \text{sign}(x_t) \in \{-1, +1\}$.

- For each $T$, estimate the rate of switching between wells (sign changes).
- Compute the distribution of first passage times to change sign (starting from $x_0 > 0$ and from $x_0 < 0$).
- Explain why at low temperature the chain can look stationary while still mixing poorly (metastability).

e) Mean Squared Jump Distance (MSJD):
Compute the mean squared jump distance $\text{MSJD} = \langle (x_{t+1} - x_t)^2 \rangle$, and plot it vs. $\Delta$ for each $T$ (using the same $\Delta$ grid as in Part c). Use this plot to justify your chosen "optimal" $\Delta$. The

average $\langle \cdot \rangle$ is approximated by a time average over $t$ such that both $t$ and $t + \tau$ lie within the recorded series.
f) Convergence diagnosis using two chains:
Run two independent chains at low temperature (e.g. $T = 0.2$) with the same $\Delta$. Start chain A at $x_0 = -1$ and chain B at $x_0 = +1$. To this end, compare histograms of chain A and chain B after the same burn-in. State whether you believe the chains have converged to the same distribution and justify your conclusion.